TITLE :

Machine Failure Prediction

Using Machine Learning

SUBJECT : Machine Learning (SE-206)

NAME - [Name of Student 1] · ROLL NUMBER - [24/SE/XXX]
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OUTLINE

1 Introduction & Motivation

Problem background and goals

2 Dataset

AI4I Predictive Maintenance — 10,000 rows

3 Methodology

6-step pipeline flowchart

4 Algorithms

LR · SVM · DT · RF · XGBoost

5 Results & Performance

Metrics comparison table

6 Conclusion & Future Work

Key findings and next steps

Motivation

- Equipment failures cost industry billions annually in unplanned downtime
- Traditional manual inspection is slow, inconsistent, and reactive
- ML enables real-time, data-driven prediction before failure occurs
- Goal: predict machine failure using sensor readings with high recall
- Imbalanced dataset (~3.4% failure) makes this a challenging ML problem

Introduction

What is Predictive Maintenance?

Predictive maintenance uses sensor data and machine learning to anticipate equipment failures before they occur — reducing downtime and costs.

Problem Statement

Given real-time sensor readings (temperature, torque, tool wear, speed), predict whether a CNC machine will fail — enabling proactive maintenance.

10,000
Data Samples

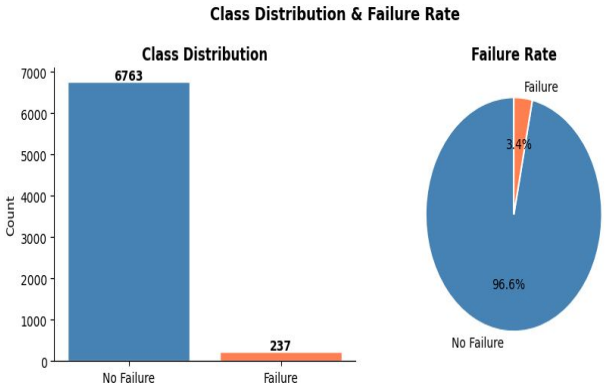
~3.4%
Failure Rate

5
ML Models

8
Eval Metrics

Dataset

Feature	Description	Type
Air temperature [K]	Ambient air temp ~300K	Numeric
Process temp [K]	Process temp ~310K	Numeric
Rotational speed [rpm]	Spindle speed ~1500rpm	Numeric
Torque [Nm]	Applied torque ~40Nm	Numeric
Tool wear [min]	Cumulative tool usage	Numeric
Type	Machine type: L / M / H	Categorical
Machine failure	Target: 0=OK, 1=Fail	Binary



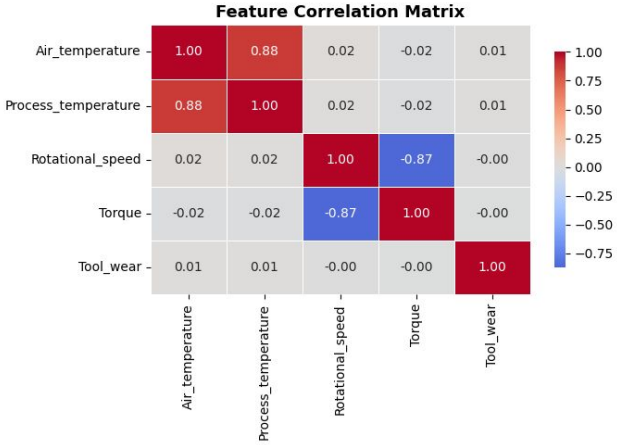
Source: AI4I 2020 Predictive Maintenance Dataset (UCI ML Repository)

Dropped: UDI, Product ID (identifiers)
Removed: TWF, HDF, PWF, OSF, RNF (leakage)

Methodology

Each step is detailed in the following slides →

Exploratory Data Analysis



Key KDE Findings

- Tool wear: failure cases concentrate in the 150–250 min range, showing clear separation from non-failure instances
- Torque: failure distribution shifts right (>50 Nm), indicating elevated torque as a strong failure indicator
- Rotational speed: failure cases peak at lower RPM values, suggesting low speed combined with high torque increases failure risk
- Air & Process temperature: KDE distributions largely overlap between classes, contributing less discriminative power individually

Key EDA Findings

- Tool wear & Torque most discriminative
- Strong Air ↔ Process temp positive correlation
- Strong Torque ↔ Rotational Speed negative correlation
- Heavy class imbalance (~3.4% failures)
- No null values or duplicate rows found

Feature Engineering

New features created from domain knowledge:

`power` *= Torque × (Rotational speed × 2π / 60)*

Mechanical power in watts — combines torque & speed

`wear_torque` *= Tool wear × Torque*

Stress accumulation — high wear under high load

`temp_diff` *= Process temp − Air temp*

Thermal delta indicates heat generation

`Type_M, Type_H` *= One-hot encoding of machine type*

L dropped as reference; bool ⟶ int conversion

Algorithms Used

Logistic Regression

Pipeline: StandardScaler → LR
class_weight='balanced', max_iter=1000

Random Forest

100 estimators
class_weight='balanced'

Support Vector Machine

Pipeline: StandardScaler → SVC
probability=True, class_weight='balanced'

XGBoost

scale_pos_weight = majority/minority ratio
eval_metric='logloss'

Decision Tree

class_weight='balanced'
random_state=42

All models evaluated with 5-fold StratifiedKFold cross-validation

Model Comparison

Model	Precision	Recall	F1	Accuracy	ROC-AUC	G-Mean	G-Measure	Specificity
Logistic Regression	0.1658	0.8396	0.2768	0.8511	0.9280	0.8454	0.3730	0.8515
SVM	0.3093	0.8650	0.4537	0.9289	0.9530	0.8967	0.5159	0.9311
Decision Tree	0.7704	0.7342	0.7500	0.9834	0.8632	0.8532	0.7511	0.9922
Random Forest	0.9829	0.7262	0.8333	0.9903	0.9665	0.8509	0.8438	0.9996
XGBoost	0.8190	0.8013	0.8076	0.9870	0.9726	0.8919	0.8089	0.9935

Hyperparameter Tuning & Overfitting Control

RandomizedSearchCV · 5-fold StratifiedKFold · Scoring: F1

Random Forest — Tuned

Search Space

- n_estimators: [100, 200, 300]
- max_depth: [3, 5, 7, 10]
- min_samples_split: [10, 20, 50]
- min_samples_leaf: [4, 8, 16]
- max_features: [sqrt, log2]
- max_samples: [0.6, 0.7, 0.8]

✔ Best Parameters Selected

- **n_estimators: 200**
- **max_depth: 10**
- min_samples_split: 20
- min_samples_leaf: 4
- max_features: log2
- max_samples: 0.7

XGBoost — Tuned

Search Space

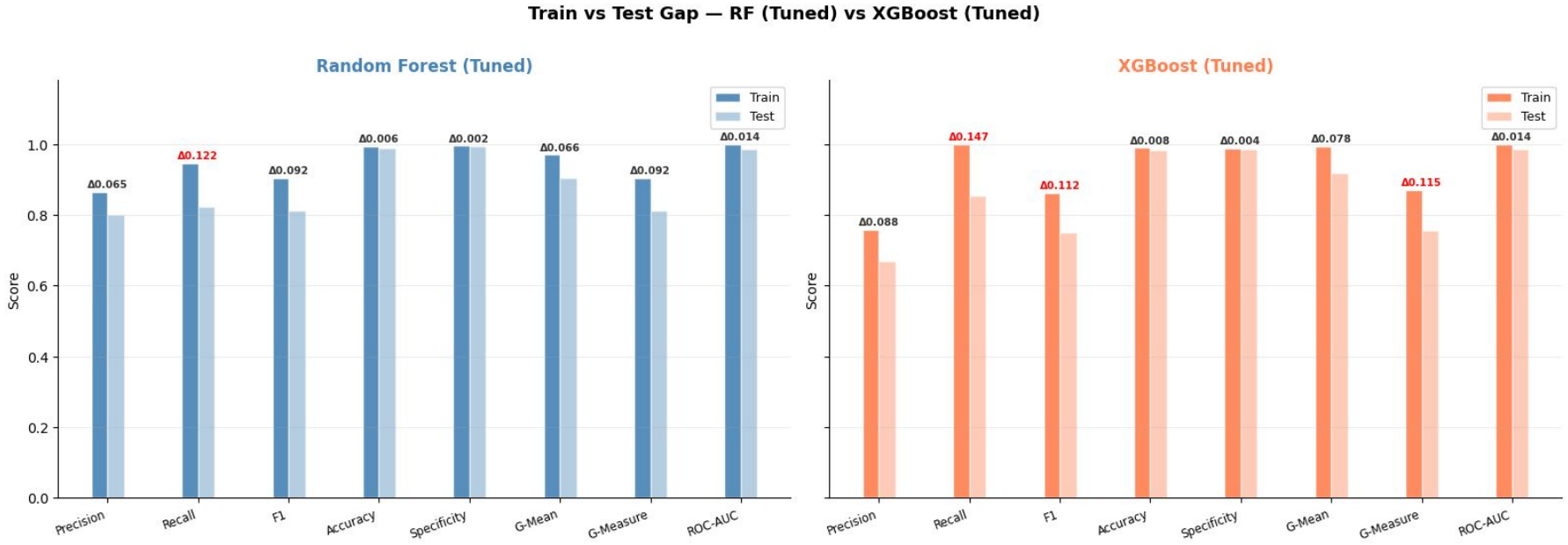
- n_estimators: [100, 200, 300]
- max_depth: [2, 3, 4]
- learning_rate: [0.01, 0.05, 0.1]
- subsample: [0.6, 0.7, 0.8]
- reg_alpha: [0.5, 1.0, 2.0] ← L1
- reg_lambda: [2.0, 5.0, 10.0] ← L2
- colsample_bytree: [0.6, 0.7, 0.8]
- min_child_weight: [10, 15 , 20]
- gamma: [0.5, 1.0 , 2.0]

✔ Best Parameters Selected

- **n_estimators: 300**
- **max_depth: 4**
- learning_rate: 0.1
- subsample: 0.8, colsample_bytree: 0.7
- reg_alpha: 2.0, reg_lambda: 10.0
- gamma: 0.5, min_child_weight: 15

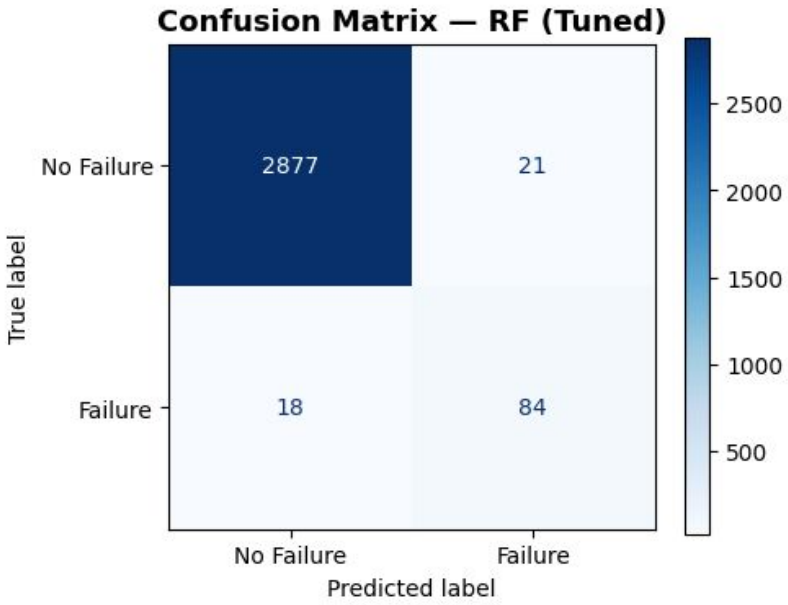
Train vs Test gap analyzed to confirm no overfitting — RF selected for better generalization

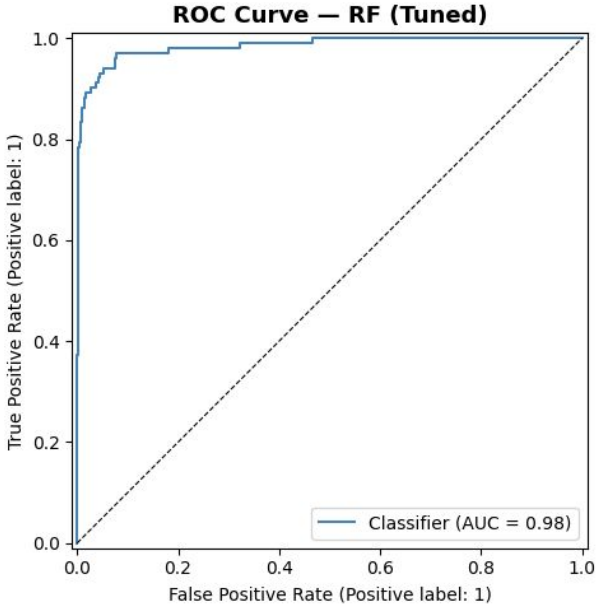
Hyperparameter Tuning & Overfitting Control



RF selected — consistently smaller train-test gap across all metrics → better generalization

Results — Best Model: RF (Tuned)

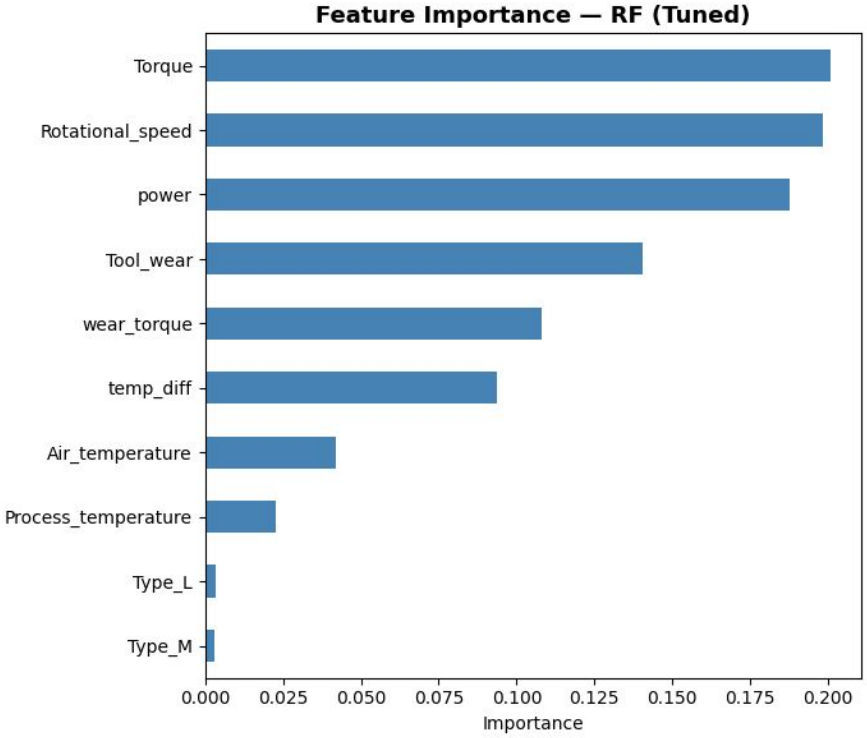


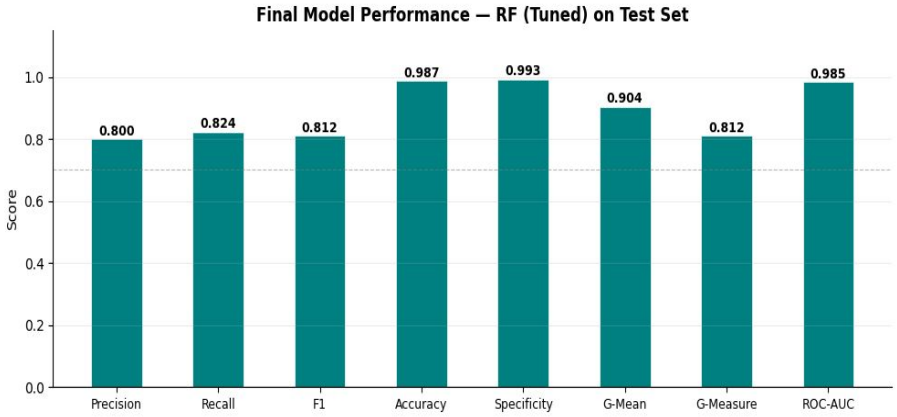


Precision: 0.8000 Recall: 0.8235 F1: 0.8116 ROC-AUC: 0.9847

High recall prioritized to minimize missed machine failures in production

Performance Analysis





Metric	Value
G-Mean	**0.9042**
G-Measure	**0.8117**
Specificity	**0.9928**
Accuracy	**0.9870**

Conclusion & Future Work

- ✔ Successfully built a machine failure prediction system using 5 ML algorithms on imbalanced data
- ✔ Random Forest (Tuned) selected as best model via RandomizedSearchCV + StratifiedKFold CV
- ✔ Feature engineering (power, wear_torque, temp_diff) improved model interpretability
- ✔ Class imbalance addressed via class_weight, scale_pos_weight, and stratified sampling

Future Scope: Apply SMOTE or ADASYN for better minority class handling · Deploy model as a real-time REST API with sensor data streaming · Explore deep learning (LSTM) on time-series sensor readings

Thank You